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System and apparatus for moving objects

Abstract

A trolley for lifting and/or moving a load on lines and an installation and method using it are disclosed. The trolley includes first and second side plates and sheave assemblies. Each sheave assembly includes a sheave axle with an axle shaft projecting from the first plate, a sheave cartridge on the axle shaft between the plates. Each sheave cartridge includes a pulley wheel rotatable on a bearing around an axis of the axle shaft. The axes of the shafts are parallel to one another. Retention fasteners extend through the second plate to secure the second and first plates through the axle shafts. An installation includes a trolley and main line extending between two fixed anchorages. In use, the main line passes between the plates of the trolley, the main line contacts a sheave or pulley wheel of the trolley whereby the trolley can travel along the main line.

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Background/Summary

(1) This application is the U.S. national phase of International Application No. PCT/GB2020/050513 filed 5 Mar. 2020, which designated the U.S. and claims priority to GB Patent Application No. 1902979.2 filed 6 Mar. 2019, the entire contents of each of which are hereby incorporated by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

(2) The present invention relates to a system and apparatus for moving objects.

Description of the Related Art

(3) A Tyrolean traverse is performed by connecting a main line between two remote locations across an expanse and placing a carriage on the main line. The carriage has a plurality of sheaves or other low-friction devices that allow it to travel along the main line. Control lines are connected to the carriage to control its movement along the main line. The carriage also has connection points to which a load can be connected. As the control lines are used to move the carriage along the main line, it is possible to move a load connected to the carriage between the remote locations. This arrangement has a variety of applications, including movement of inanimate objects, such as timber from a cut-down tree, or transport of people, either to cross a void, or to rescue an incapacitated person.

(4) EP 3 159 052 discloses a trolley for performing a Tyrolean traverse which is effective and useful. An aim of this invention is to provide a trolley that is easier to use, lighter than existing products and all components are replaceable. A trolley embodying this invention may be simpler or less expensive to manufacture without any reduction in strength and durability.

SUMMARY OF THE INVENTION

(5) To this end, from a first aspect, the invention provides a trolley for lifting and/or moving a load on one or more lines comprising: a first side plate and a second side plate and a plurality of sheave assemblies; in which each sheave assembly includes a sheave axle that includes an axle shaft which projects from the first plate, a sheave cartridge carried on the axle shaft between the plates, wherein each sheave cartridge includes a pulley wheel that can rotate on a bearing around an axis of the axle shaft, and the axes of the shafts being parallel to one another; and retention fasteners that extend through the second plate to secure the second plate to the first plate through the axle shafts.

(6) The retention fasteners and sheave axles serve to connect the plates together in a strong and solid manner. For installation, free access can be obtained to allow introduction of lines into the trolley by removal of the retention fasteners.

(7) Typical embodiments include three or more (e.g., four) sheave assemblies.

(8) Each sheave cartridge preferably includes a sheave boss that surrounds the sheave axle shaft, and which is clamped between the plates by force applied by the retention fastener. The retention fastener and sheave axle are therefore in tension, which the sheave boss provides structure in compression for reacting the tension in the sheave axle. The boss is generally fixed against rotation with respect to the plates, and preferably also with respect to an inner race of the bearing, while an outer race of the bearing is fixed for rotation with the pulley wheel.

(9) Each sheave axle is preferably secured to the first plate by a releasable fixation fastener. When secured to the first plate, part of the shaft of the sheave axle typically extends through a hole in the first plate and projects from an inner surface of the first plate, and a head of the sheave axle is typically in contact with an outer surface of the first plate. In the latter case, the head is shaped and/or dimensioned such that it cannot pass through the hole in the first plate. When secured to the first plate, axial and rotational movement of the sheave axle with respect to the plate is prevented. Note that removal of the sheave axles from the first plate is not required to gain access to the sheaves.

(10) A trolley assembly embodying the invention may comprise two (or more) interconnected trolleys as defined above, the plates of the trolleys being parallel to one another.

- (11) From a second aspect, this invention provides an installation for lifting and/or moving a load that includes a trolley or trolley assembly embodying the first aspect of the invention, a main line that extends between two fixed anchorages, and which, in use, passes between the plates of the trolley, whereby the main line makes contact with one or more sheave or sheave component of the trolley whereby the trolley can be suspended from and travel along the main line.
- (12) A typical installation further includes a control line that is connected to the trolley whereby the trolley can be drawn along the main line.
- (13) A typical installation further includes a lifting line that passes between the plates of the trolley and over two or more sheaves or sheave components of the trolley, the lifting line carrying, between the sheaves or sheave components, a device such as a pulley for connection with a load.
- (14) An alternative installation includes a second lifting line upon which a pulley assembly is carried, the pulley assembly being connected to support the trolley.
- (15) One or more fixed lines may be connected to the trolley. These can provide additional lifting capacity, for example serving as a climbing line and a backup line to enable a person to descend and ascend with respect to the trolley, for example to accompany a person being rescued.
- (16) From a third aspect, this invention provides a method of installing an installation according to the second aspect of the invention comprising: a. installing a main line between two fixed anchorages; b. removing the retention fasteners and removing the second plate; c. replacing the second plate having enclosed part of the main line between the plates and at least a pair of sheave assemblies.
- (17) Typically, prior to step c., part of a lifting line may be passed between the plates of the trolley.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Embodiments of the invention will now be described in detail, by way of example, with reference to the accompanying drawings, in which:
- (2) FIGS. 1 and 2 show a trolley embodying the invention;
- (3) FIG. 3 is a cross-section of part of the embodiment of FIGS. 1 and 2;
- (4) FIG. 4 shows the embodiment of FIGS. 1 and 2 dismantled for installation of working and lifting lines;
- (5) FIG. 5 shows an intermediate stage of the embodiment of FIGS. 1 and 2 being installed on working and lifting lines;
- (6) FIGS. 6, 7 and 8 show details of an arrangement for mounting a sheave in the embodiment of FIGS. 1 and 2;
- (7) FIGS. 9 and 10 show a retention bolt and a sheave axle in the embodiment of FIGS. 1 and 2;
- (8) FIG. 11 shows the embodiment of FIGS. 1 and 2 with one retention bolt partially unfastened;
- (9) FIG. 12 shows the embodiment of FIGS. 1 and 2 installed for use;
- (10) FIG. 13 shows the embodiment of FIGS. 1 and 2 being used in a lifting operation;
- (11) FIG. 14 shows the embodiment of FIGS. 1 and 2 installed on multiple redundant main lines;
- (12) FIGS. 15 and 16 are perspective and part cross-sectional views of a second embodiment of the invention;
- (13) FIG. 17 shows a modification to a sheave assembly of the embodiment of FIGS. 1 and 2; and
- (14) FIG. 18 shows a second modification to a sheave assembly of the embodiment of FIGS. 1 and 2.

DESCRIPTION OF THE PREFERRED EMBODIMENT

- (15) With reference to the drawings, a trolley for use in performing a Tyrolean traverse comprises first and second side plates 10, 12 of similar peripheral size and shape. The side plates 10, 12 are formed from a suitable load-bearing material such as an alloy, stainless steel or composite material.

In an orientation in which the trolley will typically be used, the plates are an approximate inverted U-shape. The plates are symmetrical about a median axis A (shown in FIG. 12) with first and second legs to opposite sides of the axis A and a recess defined between the legs. Various holes are formed in the plates, each hole, in the assembled trolley, being in alignment with a corresponding hole in the other plate.

(16) The trolley further includes four sheave assemblies, one of which is shown in detail in FIG. 3. The sheave assemblies are disposed at the vertices of a rectangle that is symmetrical about the median axis forming an upper pair of sheave assemblies and a lower pair of sheave assemblies. One of the upper sheave assemblies has been omitted from FIG. 1 to illustrate features that would otherwise be hidden.

(17) Each sheave assembly includes a sheave cartridge **20** that comprises a pulley wheel **22** carried on a rolling element bearing **24**. An outer race of the bearing **24** rotates with the pulley wheel **22** and an inner race of the bearing **24** is carried on and a sheave boss **26**. The sheave boss **26** has a central cylindrical bore centred on the axis of rotation of the pulley wheel **22**.

(18) Each sheave assembly further includes a sheave axle **30**. The sheave axle **30** has a cylindrical shaft **34** at one end of which is a head **32**. The shaft **34** is dimensioned to be a close fit within the bore of the sheave boss **26**. The head **32** extends radially from the shaft **34** and has a projecting region that has a part-circular notch **36** formed in a radially-outward-facing surface. A coaxial tapped bore **38** is formed into the shaft **34** from its end opposite that of the head **32**. A further coaxial blind hole **35** may be formed through the head **32**.

(19) Each sheave assembly includes a fixation bolt **40** and a retention bolt **2**, the purpose of which will be described below.

(20) Assembly of the trolley proceeds in stages as will now be described.

(21) A: The shaft **34** of the sheave axle **30** is inserted through a hole **44** in the first plate. The hole **44** is surrounded by a recess **46** that extends radially from the hole **44**, the head **32** of the axle **30** being received within the recess **46**. The shapes and sizes of the head **32** and the recess **46** are such that the head **32** is a close fit within the recess **46** to substantially prevent rotation of the axle **30** about its shaft **34**. When in position in the recess **46** the part-circular notch **36** is formed in an outer surface of the plate which surrounds a tapped hole **48** that passes through the first plate **10**. A fixation bolt **40** is threaded into the tapped hole **48** such that its head bears against the head **32** of the sheave axle **30** adjacent to the notch **36** and on a land **50** that partially surrounds the tapped hole **48**. Thus, the sheave axle **30** is prevented from rotating with respect to the first plate **10** by interaction between its head **32** and the recess **46** and is also prevented from sliding axially from the plate **10** by the fixation bolt **40**.

(22) B: The sheave assembly is further assembled by sliding the sheave cartridge **20** onto the shaft **34** of the sheave axle **30** such that its sheave boss **26** bears against an inner surface of the first plate **10**. This is repeated for all four sheave assemblies. This intermediate stage of assembly is shown in FIG. 4.

(23) C: The second plate **12** is then placed such that an end portion of each shaft **34** passes into a respective hole **52** in the second plate **12** until an inner surface of the second plate **12** bears against the sheave bosses **26**. Each hole **52** in which a shaft is received is counterbored from an outer surface of the second plate. The second plate **12** is retained in place by passing one of the retention bolts **42** into each of the holes such that its shaft **62** is threaded into the tapped bore **38** of the sheave axle **30**. The head **60** of the retention bolt can fit within a counterbored cavity of the hole **52** but not into the part that is not counterbored, whereby the head can clamp the second plate **12** against the sheave boss **26** as the bolt is tightened.

(24) During the stage of assembly described in the last-preceding paragraph, movement of the sheave axles **30** is prevented by the fixation bolts. This allows this stage of assembly to be performed using a single tool to tighten the retention bolts **42**: there is no need to use a tool on the axle **30** or the fixation bolt **40**.

(25) The head **60** of each retention bolt has a circumferential groove within which an O-ring **64** is located. The O-ring **64** serves two purposes: when the retention bolt **42** is tight, it is compressed between the head **60** and the counterbored cavity to resist rotation of the retention bolt **42** against unintended loosening; and it provides a visible indication of the retention bolt **42'** being insufficiently tightened, as shown at the left-hand side of FIG. **11**, by remaining visible from the exterior of the plate **12**. To enhance performance of the latter purpose, the O-ring **64** is preferably brightly coloured—for example, coloured red.

(26) In the event that a retention bolt **42** is omitted or becomes loose, the axle **30** will not be lost, because the fixation bolt **40** will prevent its removal from the first plate. The retention bolts **42** of the other sheave assemblies maintain the overall structural integrity of the trolley, and while it is believed that the strength of the trolley would not be significantly affected if one of the retention bolts were not tightened correctly or were missing entirely, this is an undesirable situation that is to be avoided if possible.

(27) Each plate **10**, **12** includes a plurality of additional pairs of circular mounting holes **90**. Each pair of mounting holes **90** can receive a removable pin **92**, which locks into place in the holes until it is released by a user-operated trigger.

(28) When not in use, each retention bolt can be inserted into a blind hole e.g. **35** in the head **32** of the sheave axle **30**.

(29) Assembly of a complete installation for moving items using the trolley describe above will now be described.

(30) In addition to the trolley, the installation comprises several lines formed from ropes or cables. These are: “main line”: a line that bears the majority of the load and that allows the trolley to move in a transverse direction across an expanse to be crossed. The main line is typically fixed at opposite sides of the expanse during use. “lifting line”: this is also load-bearing but is controllable by the user to raise or lower the load independently of the main line. The lifting line may be fixed at one side and adjustable in length at the other side of the expanse during use, or it may be adjustable at both sides. “control line”: causes movement and control of the trolley back and forth along the main line. It is fixed to the trolley. This line could be controlled by the user using a device that provides mechanical advantage (e.g., a pulley system) or may be under powered control (e.g., using a winch).

(31) A typical sequence of assembly is as follows: The main **80** line and the lifting line **82** are installed across the expanse. The trolley is partially assembled as described in stages A and B above, to achieve the arrangement shown in FIG. **5**. The first plate **10** is placed adjacent to the main line **80** and the lifting line **82** such that the lines pass between the upper and lower pairs of sheave assemblies, and the second plate **12** is then installed as described in paragraph C above. A pulley block **86** is attached to the lifting line **82** between the lower pair of sheave assemblies. A control line is connected to a pin **92** inserted into a pair of mounting holes **90**.

(32) Slackening or loosening the lifting line **82** causes a load **L** attached to the pulley block **86** to be lowered or raised with respect to the trolley. As shown in FIG. **12**, in a raised position, the pulley block **86** is received within the recess between the legs of the trolley. When the pulley block **86** is in this position, the lifting line passes **82** through the trolley along an approximately straight path, which minimises friction arising from movement of the trolley.

(33) In installations where the main line **82** is sloping, one control line **84** can be used to pull the trolley in an upward direction on the main line **80** or released in a controlled manner to cause the trolley to move in a downward direction along the main line **80** under the action of gravity. Where there is no overall slope, two control lines **84**, **84'** may be provided to pull the trolley in opposite directions along the main line **80**.

(34) The arrangement of FIG. **14** uses the same trolley described above. Series redundancy is created by providing an additional main line **88**. An additional pulley assembly **96** is carried on the additional main line **88** and connected to the trolley using a pin **92** inserted into a pair of mounting

holes **90** towards the top of the plates **10, 12**. In that way, should the main line **82** fail, the additional main line serves as a backup to prevent loss of a load being carried. This arrangement can be used with a single control line **84** or two control lines **84, 84'** as needed.

(35) Also shown in this embodiment are additional lines **98** that can be used to carry a load. These are fixed lines that are each secured to the trolley using a pin **92** inserted into a pair of mounting holes **90** towards the bottom of the plates **10, 12**. A typical use for these fixed lines **98** would be to allow a person to accompany a load, such as an injured person, being lifted on the pulley block **86**. In such an application, one of the fixed lines **98** would be used as a climbing line upon which a climber could move using an ascender, and the other fixed line **98** would be used as a back-up line to protect the climber against failure of the climbing line and associated apparatus.

(36) A trolley as described above could be paired, as shown in FIGS. **15** and **16**, with another identical trolley to create a system which has theoretically twice the strength and lifting capacity as that described above, but most importantly for some users gives redundancy in the system and a backup device through a parallel redundancy system. Using such a trolley, each line and piece of equipment being part of an installation can be duplicated to provide redundancy and therefore protect against failure of multiple components in the installation. Two trolleys are joined together using long bolts **110** and spacers through the first and second plates **10, 12**.

(37) While stage B of the assembly sequence described above is being performed, the sheave cartridges **20** have been slid onto the shafts **34** of the sheave axles **30**, but they are not retained there, which means that they could slide off and potentially become lost. To minimise the risk of this happening, in a modification of the above-described embodiment shown in FIG. **17**, a circumferential groove is formed on the shafts **34** of the sheave axles **30** and an O-ring **120** is retained in the groove. The sheave boss **26** is gripped by the O-ring **120** with sufficient force that it will not fall off the shaft **34** accidentally under the weight of the sheave assembly, but such that it can be removed using light manual effort when required.

(38) A further alternative configuration of the sheave assembly is shown in FIG. **18**. In this arrangement, there is no sheave boss **26**, and the inner race of the bearing **24** is carried directly upon the shaft **234** of the sheave axle **230**. As compared with the arrangement described above, the shaft **234** of the sheave axle **230** is of larger diameter so that the size of the bearing **24** can remain unchanged. Axial location of the bearing **24** on the axle **230** is provided by annular abutments **240, 242** that project from the first and second plates **210, 212**. The abutments **240, 242** closely surround the shaft **234** of the sheave axle **230**, and make contact with opposite side faces of the inner race of the bearing **24** to prevent it moving axially along the shaft **234**, and thereby preventing axial movement of the pulley wheel **22**.

(39) An O-ring **244** is located in a groove formed on the shaft **234** of the sheave axle **230** at a position that is radially inward of the inner race when the bearing **24** is in its operational position on the shaft **234**. As with the previous embodiment, the inner race of the bearing **24** is gripped by the O-ring **244** with sufficient force that it will not fall off the shaft **234** accidentally, but such that it can be removed manually when required.

Claims

1. A trolley for lifting and/or moving a load on one or more lines comprising: a first side plate and a second side plate; and a plurality of sheave assemblies in which each sheave assembly includes a sheave axle that includes an axle shaft which projects from the first plate and passes into a hole in the second plate, wherein the sheave axle is in tension between the plates, and a sheave cartridge carried directly on the axle shaft between the plates, wherein each sheave cartridge includes: a pulley wheel rotatable on a bearing around an axis of the axle shaft; and a removable retention fastener that extends through the hole in the second plate to engage directly with the axle shaft to removably secure the second plate to the first plate via the axle shaft, wherein the axes of the axle

shafts of the plurality of sheave assemblies are parallel to one another, wherein the sheave axle is secured to the first plate by the releasable fixation fastener with axial and rotational movement of the sheave axle with respect to the first plate being prevented, and wherein removal of the sheave axle from the first plate is not required to gain access to each sheave assembly.

2. The trolley according to claim 1 that includes three or more sheave assemblies.

3. The trolley according to claim 2, wherein each sheave cartridge includes a sheave boss that surrounds the sheave axle shaft, and which is clamped between the plates by force applied by the retention fastener.

4. The trolley according to claim 1, wherein each sheave cartridge includes a sheave boss that surrounds the sheave axle shaft, and which is clamped between the plates by force applied by the retention fastener.

5. The trolley according to claim 3, wherein the sheave boss provides a structure that can be loaded in compression to counter react the tension force in the sheave axle.

6. The trolley according to claim 4, wherein the boss is fixed against rotation with respect to the plates, and with respect to an inner race of the bearing of the sheave cartridge, while an outer race of the bearing is fixed for rotation with the pulley wheel.

7. The trolley according to claim 1, wherein part of the shaft of the sheave axle extends through a hole in the first plate and projects from an inner surface of the first plate.

8. The trolley according to claim 1, wherein a head of the sheave axle is in contact with an outer surface of the first plate.

9. The trolley according to claim 8, wherein the head is shaped and/or dimensioned such that the head cannot pass through a hole in the first plate.

10. A trolley assembly, the trolley assembly comprising two trolleys according to claim 1, the two trolleys being interconnected with the plates of the trolleys being parallel to one another.

11. An installation for lifting and/or moving a load, comprising a trolley according to claim 1, and a main line that extends between two fixed anchorages, and which, in use, passes between the plates of the trolley, whereby the main line makes contact with one or more sheave or sheave component of the trolley whereby the trolley can be suspended from and travel along the main line.

12. The installation according to claim 11, further comprising a control line that is connected to the trolley whereby the trolley can be drawn along the main line.

13. The installation according to claim 11, further comprising a lifting line that passes between the plates of the trolley and over two or more sheaves or sheave components of the trolley, the lifting line carrying, between the sheaves or sheave components, a device for connection with a load.

14. The installation according to claim 13, wherein the device for connection with a load includes a pulley.

15. The installation according to claim 11, further comprising a second lifting line upon which a pulley assembly is carried, the pulley assembly being connected to support the trolley.

16. The installation according to claim 11, further comprising one or more fixed lines connected to the trolley.

17. A method of installing an installation according to claim 11, the method comprising: installing the main line between two fixed anchorages; removing the retention fasteners and removing the second plate; and replacing the second plate having enclosed part of the main line between the plates and at least a pair of sheave assemblies.

18. The method according to claim 17 further comprising, prior to the step of replacing the second plate, passing part of a lifting line between the plates of the trolley.

19. The trolley according to claim 1, wherein the bearing has an outer race and an inner race, the inner race mounted on the axle shaft, and wherein the pulley wheel is carried on the outer race of the bearing around an axis of the axle shaft, the outer race rotating with the pulley wheel and the inner race being carried on a sheave boss.
